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WEEK 9

BAGAIAN 1

Pertama lakukan traceroute ke gaia.cs.umass.edu

Gunakan filter pada Wireshark:

- udp || icmp
- ip.src==192.168.86.61 && ip.dst==128.119.245.12 && udp && !icmp
- ip.dst==192.168.86.61 && icmp

Hasil

Wireshark packet capture analysis showing network traffic details, including packet lists, packet details, and packet bytes.

Packet List (Top):

No.	Time	Source	Destination	Protocol	Length	Info
2	4.866867	192.168.1.100	192.168.1.1	SSDP	174	M-SEARCH * HTTP/1.1
3	4.869147	192.168.1.100	192.168.1.1	SSDP	175	M-SEARCH * HTTP/1.1
4	5.365336	192.168.1.100	192.168.1.1	SSDP	174	M-SEARCH * HTTP/1.1
5	5.364799	192.168.1.100	192.168.1.1	SSDP	175	M-SEARCH * HTTP/1.1
6	5.864428	192.168.1.100	192.168.1.1	SSDP	174	M-SEARCH * HTTP/1.1
7	5.865461	192.168.1.100	192.168.1.1	SSDP	175	M-SEARCH * HTTP/1.1
8	6.010805	192.168.1.102	128.59.23.100	ICMP	98	Echo (ping) request id=0x0300, seq=20483/849, ttl=1 (no response found)
9	6.176826	192.168.1.102	128.59.23.100	ICMP	98	Echo (ping) request id=0x0300, seq=20739/849, ttl=2 (no response found)
10	6.188629	192.168.1.102	128.59.23.100	ICMP	98	Echo (ping) request id=0x0300, seq=20995/850, ttl=3 (no response found)
11	6.202957	192.168.1.102	128.59.23.100	ICMP	98	Echo (ping) request id=0x0300, seq=21251/851, ttl=4 (no response found)
12	6.208597	192.168.1.102	128.59.23.100	ICMP	98	Echo (ping) request id=0x0300, seq=21507/852, ttl=5 (no response found)
13	6.222455	192.168.1.102	128.59.23.100	ICMP	98	Echo (ping) request id=0x0300, seq=21507/852, ttl=5 (no response found)
14	6.238695	192.168.1.102	128.59.23.100	ICMP	98	Echo (ping) request id=0x0300, seq=21507/852, ttl=5 (no response found)
15	6.257672	192.168.1.102	128.59.23.100	ICMP	98	Echo (ping) request id=0x0300, seq=21507/852, ttl=5 (no response found)
16	6.258750	192.168.1.102	128.59.23.100	ICMP	98	Echo (ping) request id=0x0300, seq=21507/852, ttl=5 (no response found)
17	6.269817	12.125.47.49	192.168.1.102	ICMP	98	Echo (ping) request id=0x0300, seq=21507/852, ttl=5 (no response found)

Packet Details (Middle):

- Frame 2: Packet, 174 bytes on wire (1392 bits), 174 bytes captured (1392 bits) on interface eth0
- Ethernet II, Src: Intel, Dst: LinksysGroup, Seq: 73 (00:06:25:da:af:73)
- Internet Protocol Version 4, Src: 192.168.1.100, Dst: 192.168.1.1
- User Datagram Protocol, Src Port: 30955, Dst Port: 1900
- Simple Service Discovery Protocol

Packet Bytes (Bottom):

0000 00 06 25 da af 73 00 04 22 52 26 23 00 00 45 00 ... s s R#4: E

0010 00 00 c6 1a 00 00 01 11 6f 7d c0 a8 01 64 c0 a8 ... o) d

0020 01 01 78 eb 07 6c 00 8c 96 1c 4d 2d 53 45 41 52 ... x 1 M-SEAR

0030 43 48 20 2a 20 48 54 54 50 2f 31 2e 31 0d 0a 48 ... CH * HT P/1.1 H

0040 4f 53 54 3a 20 32 33 39 2e 32 35 35 2e 32 35 35 ... OST: 239 255.255

0050 2a 32 35 30 3a 31 39 30 30 0a 4d 41 4a 3a 20 ... 250:190 0 RAN:

0060 22 73 73 64 70 3a 64 69 73 63 6f 76 65 72 22 0d ... "ssdp:discover"

0070 0a 4d 58 3a 20 32 0d 0a 53 54 3a 20 75 72 6e 3a ... RK: 2 ST: urn:

0080 73 63 68 65 6d 61 73 2d 75 70 6e 70 2d 6f 72 67 ... scheme: upnp.org

0090 3a 73 65 72 76 69 63 65 3a 57 41 4e 49 50 43 6f ... iservice:WANIPCo

00a0 6e 6e 65 63 74 69 6f 6e 3a 31 0d 0a 0d 0a ... nnection:11 ...

Packet List (Bottom):

No.	Time	Source	Destination	Protocol	Length	Info
44	1.865637	192.168.86.61	128.119.245.12	UDP	70	64928 → 33435 Len=28
48	1.874816	192.168.86.61	128.119.245.12	UDP	70	64928 → 33436 Len=28
50	1.875401	192.168.86.61	128.119.245.12	UDP	70	64928 → 33437 Len=28
52	1.876720	192.168.86.61	128.119.245.12	UDP	70	64928 → 33438 Len=28
56	1.885567	192.168.86.61	128.119.245.12	UDP	70	64928 → 33439 Len=28
58	1.889002	192.168.86.61	128.119.245.12	UDP	70	64928 → 33440 Len=28
60	1.892656	192.168.86.61	128.119.245.12	UDP	70	64928 → 33441 Len=28
62	1.897036	192.168.86.61	128.119.245.12	UDP	70	64928 → 33442 Len=28
64	1.928173	192.168.86.61	128.119.245.12	UDP	70	64928 → 33443 Len=28
67	1.940279	192.168.86.61	128.119.245.12	UDP	70	64928 → 33444 Len=28
69	1.951481	192.168.86.61	128.119.245.12	UDP	70	64928 → 33445 Len=28
71	1.965335	192.168.86.61	128.119.245.12	UDP	70	64928 → 33446 Len=28
73	1.975799	192.168.86.61	128.119.245.12	UDP	70	64928 → 33447 Len=28
75	1.991739	192.168.86.61	128.119.245.12	UDP	70	64928 → 33448 Len=28
77	2.008887	192.168.86.61	128.119.245.12	UDP	70	64928 → 33449 Len=28
79	2.025822	192.168.86.61	128.119.245.12	UDP	70	64928 → 33450 Len=28
81	2.045745	192.168.86.61	128.119.245.12	UDP	70	64928 → 33451 Len=28
83	2.063103	192.168.86.61	128.119.245.12	UDP	70	64928 → 33452 Len=28
85	2.081361	192.168.86.61	128.119.245.12	UDP	70	64928 → 33453 Len=28
87	2.104137	192.168.86.61	128.119.245.12	UDP	70	64928 → 33454 Len=28
89	2.122238	192.168.86.61	128.119.245.12	UDP	70	64928 → 33455 Len=28

Packet Details (Bottom):

- Frame 44: Packet, 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface en0, id 0
- Ethernet II, Src: Apple, Dst: Google, Seq: 89 (0e:8d:89:0e:c8)
- Internet Protocol Version 4, Src: 192.168.86.61, Dst: 128.119.245.12
- User Datagram Protocol, Src Port: 64928, Dst Port: 33435
- Data (28 bytes)

Packet Bytes (Bottom):

0000 3c 28 6d 89 0e c8 78 4f 43 98 d9 27 08 00 45 00 ... <(m...C...E

0010 00 3f fd a1 00 00 01 11 2f aa c0 a8 56 3d 80 77 ... 8.../...Vw w

0020 f5 0c fd a0 82 9b 00 24 f2 ff 00 00 00 00 00 00 ... \$...Vw w

0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

0040 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

0050 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

BAGIAN 2

Langkah

Jalankan traceroute dengan ukuran 3000 byte.

Hilangkan filter, urutkan paket berdasarkan kolom Time.

Cari paket UDP besar (Len $\approx$ 2972).

Perhatikan fragmen IPv4 dengan Identification sama, Offset berbeda, dan MF flag nya.

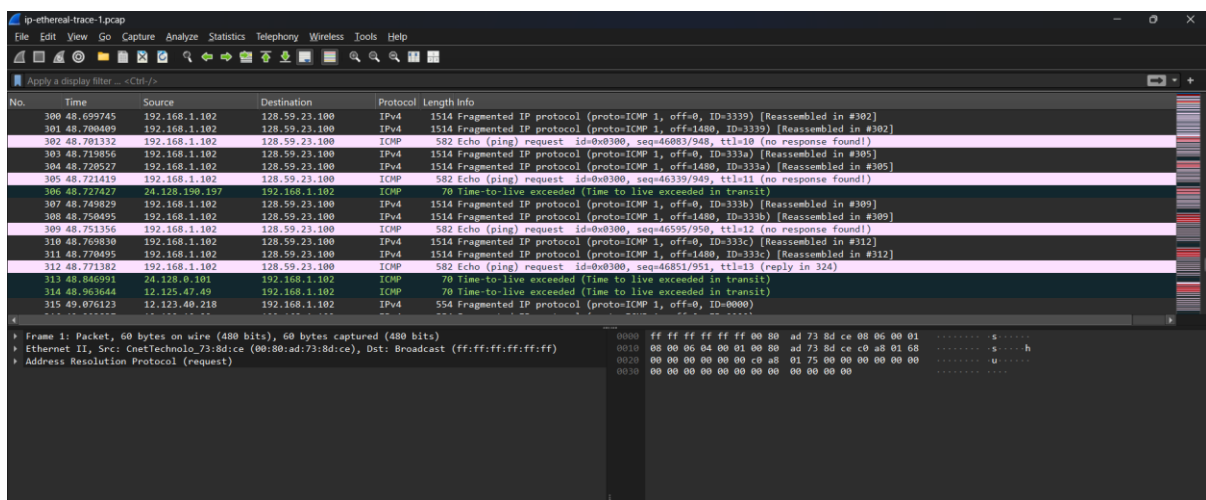
Hasil

Paket UDP besar (2972 byte) terfragmentasi.

Fragmen pertama: Offset = 0, MF = 1.

Fragmen kedua: Offset = 1480, MF = 0.

Semua fragmen punya Identification sama.



BAGIAN 3

Langkah

Buka file ip-wireshark-trace2-1.pcapng.

Gunakan filter: frame.number < 300.

Amati paket ke-20 (t=3.814489).

Hasil

Pake ke 20 berisi ICMP Echo Request (ping request) Dikirim dari komputer Menuju server tujuan (128.59.23.100 / gaia.cs.umass.edu) Dengan TTL = 7

Header IPv6 menampilkan Traffic Class, Flow Label, Payload Length, Next Header, Hop Limit.

Alamat sumber dan tujuan berupa IPv6.

ip-etherreal-trace-1.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

frame.number < 300

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	CnetTechnolo_73:8d:...	Broadcast	ARP	60	Who has 192.168.1.117? Tell 192.168.1.104
2	4.866867	192.168.1.100	192.168.1.1	SSDP	174	M-SEARCH * HTTP/1.1
3	4.868147	192.168.1.100	192.168.1.1	SSDP	175	M-SEARCH * HTTP/1.1
4	5.363536	192.168.1.100	192.168.1.1	SSDP	174	M-SEARCH * HTTP/1.1
5	5.364799	192.168.1.100	192.168.1.1	SSDP	175	M-SEARCH * HTTP/1.1
6	5.864428	192.168.1.100	192.168.1.1	SSDP	174	M-SEARCH * HTTP/1.1
7	5.865461	192.168.1.100	192.168.1.1	SSDP	175	M-SEARCH * HTTP/1.1
8	6.163845	192.168.1.102	128.59.23.100	ICMP	98	Echo (ping) request id=0x0300, seq=20483/848, ttl=1 (no response found!)
9	6.168216	192.168.229.81	128.59.23.100	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
10	6.188629	192.168.1.102	128.59.23.100	ICMP	98	Echo (ping) request id=0x0300, seq=20739/849, ttl=2 (no response found!)
11	6.202957	24.218.0.153	192.168.1.102	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
12	6.208597	192.168.1.102	128.59.23.100	ICMP	98	Echo (ping) request id=0x0300, seq=20995/850, ttl=3 (no response found!)
13	6.214568	24.218.190.107	192.168.1.102	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
14	6.238695	192.168.1.102	128.59.23.100	ICMP	98	Echo (ping) request id=0x0300, seq=21251/851, ttl=4 (no response found!)
15	6.257672	24.128.0.101	192.168.1.102	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
16	6.258750	192.168.1.102	128.59.23.100	ICMP	98	Echo (ping) request id=0x0300, seq=21507/852, ttl=5 (no response found!)

Frame 1: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits)

Ethernet II, Src: CnetTechnolo_73:8d:ce (00:80:ad:73:8d:ce), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

Address Resolution Protocol (request)

0000 ff ff ff ff ff ff 00 80 ad 73 8d ce 08 06 00 01 s
0010 08 00 06 04 00 01 00 80 ad 73 8d ce c0 a8 01 68 sh
0020 00 00 00 00 00 00 c0 a8 01 75 00 00 00 00 00 u
0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00